

First and Follow

During top down parsing, FIRST and FOLLOW allow us to choose which production to apply, based on the next input (look a head) symbol.

Computation of FIRST:

FIRST(X) is the set of terminals that can appear as the **first symbol** in strings derived from X.

RULES FOR FIRST

Rule 1: If X is a terminal

$$\text{FIRST}(X) = \{X\}$$

$$\text{FIRST}(a) = \{a\}$$

$$\text{FIRST}(\text{id}) = \{\text{id}\}$$

Rule 2: If $X \rightarrow \epsilon$

$$\epsilon \in \text{FIRST}(X)$$

$$A \rightarrow \epsilon$$

$$\text{FIRST}(A) = \{\epsilon\}$$

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Rule 3: If $X \rightarrow Y_1 Y_2 Y_3 \dots$

Steps:

1. Add $\text{FIRST}(Y_1) - \{\epsilon\}$
2. If Y_1 has ϵ , go to Y_2
3. Continue until a symbol does NOT produce ϵ
4. If all produce $\epsilon \rightarrow$ add ϵ to $\text{FIRST}(X)$

Grammar:

$S \rightarrow AB$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

Step 1: $\text{FIRST}(B)$

- $B \rightarrow b$
- $\text{First}\{B\} = \{B\}$

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Step 2: FIRST(A)

- $A \rightarrow a \mid \varepsilon$
- $\text{First } \{A\} = \{a, \varepsilon\}$

Step 3: FIRST(S)

$S \rightarrow AB$

- $\text{FIRST}(A) = \{a, \varepsilon\} \rightarrow \text{add } a$
- Since ε exists, check $B \rightarrow \text{add } b$

$\text{FIRST}(S) = \{a, b\}$

First

S -> ABC/ ghi / jkl

A -> a/b/c

B-> b

D -> d

First

S -> **ABC** / ghi / jkl {a,b,c, g,j}

A -> a/b/c {a, b,c}

B -> b {b}

D -> d {d}

S -> **ABC**

A -> a/ b / ϵ

B -> c/ d / ϵ

C -> e/ f/ ϵ

E -> **TE'**

E' -> ***TE'** / ϵ

T -> **FT'**

T' -> ϵ / +**FT'**

F -> id / (E

First

S -> **ABC/ ghi / jkl** {a,b,c, g,j}

A -> **a/b/c** {a, b,c}

B -> **b** {b}

D -> **d** {d}

S -> **ABC** {a,b, c,d, e,f , ϵ }

A -> **a/ b / ϵ** {a,b, ϵ }

B -> **c/ d / ϵ** {c,d, ϵ }

C -> **e/ f/ ϵ** {e,f, ϵ }

E -> **TE'**

E' -> ***TE' / ϵ**

T -> **FT'**

T' -> **ϵ / +FT'**

F -> **id / (E**

First

S -> **ABC/ ghi / jkl** {a,b,c, g,j}

A -> **a/b/c** {a, b,c}

B -> **b** {b}

D -> **d** {d}

S -> **ABC** {a,b, c,d, e,f , ϵ }

A -> **a/ b / ϵ** {a,b, ϵ }

B -> **c/ d / ϵ** {c,d, ϵ }

C -> **e/ f/ ϵ** {e,f, ϵ }

E -> **TE'** {id , (}

E' -> ***TE' / ϵ** {*, ϵ }

T -> **FT'** {id, (}

T' -> **ϵ / +FT'** { ϵ , +}

F -> **id / (E** {id, (}

Follow

FOLLOW(A) is the set of terminals that can appear **immediately after A** in some derivation.

Rule 1: Start Symbol

If S is start symbol:

$$\text{FOLLOW}(S) = \{\$ \}$$

Rule 2: $A \rightarrow \alpha B \beta$

If B is followed by β :

Add:

$$\text{FIRST}(\beta) - \{\epsilon\} \text{ to } \text{FOLLOW}(B)$$

Rule 3: $A \rightarrow \alpha B$

If B is at the end:

$$A \rightarrow \alpha B$$

Then:

Add FOLLOW(A) to FOLLOW(B)

Follow

Grammar

$S \rightarrow AB$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

$\text{Follow}(S) = \{\$ \}$

$\text{Follow}(A) = \text{First}(B) = b$

$\text{Follow}(B) = \{\$ \}$

Follow

Grammar

$S \rightarrow AB$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

$\text{Follow}(S) = \{\$, \}$

$\text{Follow}(A) = \text{First}(B) = b$

$\text{Follow}(B) = \{\$, \}$

Find follow of

$S \rightarrow ABC / ghi / jkl \quad \{a, b, c, g, j\}$

$A \rightarrow a/b/c \quad \{a, b, c\}$

$B \rightarrow b \quad \{b\}$

$D \rightarrow d \quad \{d\}$

Follow

Grammar

$S \rightarrow AB$

$A \rightarrow a \mid \epsilon$

$B \rightarrow b$

$\text{Follow}(S) = \{\$\}$

$\text{Follow}(A) = \text{First}(B) = b$

$\text{Follow}(B) = \{\$\}$

Find follow of

$S \rightarrow ABC / ghi / jkl \quad \{a, b, c, g, j\}$

$A \rightarrow a/b/c \quad \{a, b, c\}$

$B \rightarrow b \quad \{b\}$

$D \rightarrow d \quad \{d\}$

Non-Terminal	FIRST	FOLLOW
S	{ a, b, c, g, j }	{ \$ }
A	{ a, b, c }	{ b }
B	{ b }	{ c } (if C is a terminal) { d } (if C was a typo for D)
D	{ d }	{ } (does not appear on the RHS)

Follow

S $\rightarrow$ ABC

A $\rightarrow$ a / b / ϵ

B $\rightarrow$ c / d / ϵ

C $\rightarrow$ e / f / ϵ

Follow

S → ABC

A → a / b / ϵ

B → c / d / ϵ

C → e / f / ϵ

Non-Terminal	FIRST	FOLLOW
S	{ a, b, c, d, e, f, ϵ }	{ \$ }
A	{ a, b, ϵ }	{ c, d, e, f, \$ }
B	{ c, d, ϵ }	{ e, f, \$ }
C	{ e, f, ϵ }	{ \$ }

Follow

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \varepsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \varepsilon$$

$$F \rightarrow \text{id} \mid (E)$$

Find first and follow

$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow +TE' \mid \varepsilon \\ T &\rightarrow FT' \\ T' &\rightarrow *FT' \mid \varepsilon \\ F &\rightarrow \text{id} \mid (E) \end{aligned}$$
$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow +TE' \mid \varepsilon \\ T &\rightarrow FT' \\ T' &\rightarrow *FT' \mid \varepsilon \\ F &\rightarrow \text{id} \mid (E) \end{aligned}$$

First

First & Follow

	FIRST	FOLLOW
$E \rightarrow TE'$	{id,(}	{\$,)}
$E' \rightarrow +TE' \mid \varepsilon$	{+, ε }	{\$,)}
$T \rightarrow FT'$	{id,(}	{+,\$,)}
$T' \rightarrow *FT' \mid \varepsilon$	{*, ε }	{+,\$,)}
$F \rightarrow id \mid (E)$	{id,(}	{*,+,\$,)}

Find first and follow for

S $\rightarrow$ ABCDE

A $\rightarrow$ a / ϵ

B $\rightarrow$ b / ϵ

C $\rightarrow$ c

D $\rightarrow$ d / ϵ

E $\rightarrow$ e / ϵ

Find first and follow for

$S \rightarrow ABCDE$

$A \rightarrow a / \epsilon$

$B \rightarrow b / \epsilon$

$C \rightarrow c$

$D \rightarrow d / \epsilon$

$E \rightarrow e / \epsilon$

	FIRST	FOLLOW
$S \rightarrow ABCDE$	$\{a, b, c\}$	$\{\$ \}$
$A \rightarrow a \mid \epsilon$	$\{a, \epsilon\}$	$\{b, c\}$
$B \rightarrow b \mid \epsilon$	$\{b, \epsilon\}$	$\{c\}$
$C \rightarrow c$	$\{c\}$	$\{d, e, \$ \}$
$D \rightarrow d \mid \epsilon$	$\{d, \epsilon\}$	$\{e, \$ \}$
$E \rightarrow e \mid \epsilon$	$\{e, \epsilon\}$	$\{\$ \}$